

DEEPFAKE IDENTITY DETECTION: A Convolutional Neural Network Approach

A Project Work

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by

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CERTIFICATE

This is to certify that this project work entitled **Deepfake Identity Detection: A Convolutional Neural Network Approach** is a bonafide record of work carried out by **Malavika S(233143010015)** in partial fulfillment of the requirements for the award of the degree of **Integrated M.Sc Computer Science in Artificial Intelligence and Machine Learning** of Mahatma Gandhi University in the Department of Computer Science, Bharata Mata College, Thrikkakara during the academic year 2025-26.

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DECLARATION

I declare that the work presented in this project “Deepfake Identity Detection: A Convolutional Neural Network Approach “, portrays my ideas and where other’s ideas or words are included is adequately cited and referenced the original source. I also declare that I have adhered to all the principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/source in my submission. I understand that any violation of the above will cause a disciplinary action by the university and can also evoke personal actions from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

In recent years, the widespread availability of digital image editing tools and the rapid development of artificial intelligence-based image generation technologies have made it increasingly difficult to determine the authenticity of visual content. Deepfakes, AI-generated images, and altered media pose substantial privacy, security and information integrity issues surrounding information that is shared online. The main research areas of this project are the development of an automatic real or fake image identification system by machine learning.

The proposed model employs Transfer Learning utilizing a lightweight MobileNetV2 convolutional neural network architecture, chosen for its computational efficiency and effectiveness in image classification tasks. Optimized processing power for the system is achieved by first using an initial "Security Gate" to detect and filter out images that are not of a human face using OpenCV Haar Cascades before the images are analyzed by AI. These input images are then preprocessed by normalization and resizing to boost the model performance and generalization. The MobileNetV2 model learns and filters out unique visual details that enable the distinction between a real person and fake one, using "Strict-Mode" logic with a confidence threshold of 0.75, to strictly reduce false positives. The metrics typically used for evaluating model performance include accuracy, precision, recall, and F1-score, ensuring the reliability and robustness of the model.

Furthermore, a user-friendly web-based interface is created using Python and a web development framework called Streamlit, allowing the user to upload an image and get an authenticity prediction immediately. The outcomes of this project demonstrate the potential of lightweight deep learning approaches in accurately detecting manipulated images, thereby helping to combat misinformation, digital fraud, and unethical media practices.

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List of Abbreviations

AI Artificial Intelligence

CNN Convolutional Neural Network

GAN Generative Adversarial Network

ML Machine Learning

OpenCV Open Source Computer Vision Library

PIL Python Imaging Library

UI User Interface

TP True Positive

TN True Negative

FP False Positive

FN False Negative

GAP Global Average Pooling

ROI Region of Interest

Chapter 1

INTRODUCTION

In the field of digital security and cybersecurity, the spread of hyper-realistic AI-generated imagery continues to pose a major risk to identity verification globally. This phenomenon, that can be mainly attributed to the progress of Generative Adversarial Networks (GANs) such as StyleGAN, enables the generation of fully synthetic, hyper-realistic human faces. To address this challenge, innovative approaches are needed to improve detection accuracy and to simplify the verification processes on standard computing devices. Here, applying deep learning techniques seems a promising direction. This thesis focuses on the investigation and realization of a Convolutional Neural Network (CNN)-based deepfake detection approach, with a particular emphasis on a lightweight architecture designed for digital portraits.

This project deals with the application of deep learning techniques to detect deepfakes from normal facial images. With a large data set of 140,000 images, it is possible to train machine learning models to identify the subtle spatial artifacts and pixel level inconsistencies that are characteristic of GAN generated media . This has the potential to revolutionize digital identity verification . The main goal of this project is to assess and deploy a computationally efficient model i.e MobileNetV2 for classifying the images for deepfake detection. This work includes the design, tuning and evaluation of this classifier to achieve high performance in terms of accuracy, precision, recall and security optimization.

The significance of this study is that it can enhance the security features on digital platforms, reducing the need for manual moderation, and making it possible to verify the identity automatically in real-time. This research intends to help

develop reliable and automated tools for digital security by designing a lightweight classifier with high accuracy and a strict face-validation gate so that digital trust can be restored and the workload of human authenticators can be reduced.

1.1 Motivation

The recent breakthroughs in deep learning and generative models have made it possible to generate hyper-realistic deepfake videos that can be used to influence public perception and propagate misinformation. As deepfake generation tools become more accessible, they have also raised serious ethical, social and security concerns. Malicious actors can exploit this technology for political manipulation, fraud, cyberbullying and defamation.

Detecting deepfakes is a challenging task as the generation techniques evolve continuously to eliminate visual artifacts and generate media with high fidelity to real ones. But, current deepfake detection models are often computationally heavy, limiting their accessibility for rapid, real-time deployment on standard consumer hardware. The motivation of this project is to create a fast, accessible and secure detection framework. Through the use of MobileNetV2's computational efficiency and a rigorous security threshold tailored to avoid AI-generated fraud in safety-critical settings, we seek to improve deepfake detection performance and support the larger objective of protecting media integrity in the digital era.

1.2 Problem Statement and Objectives

1.2.1 Problem Statement

The Deepfake Identity Detection Project exists to create a computerized method with high precision and low resource usage because identifying human faces created by machine intelligence is necessary. Current deepfake detection models are very heavy for computers and demand large server rooms and high-cost processing units for image analysis. Security is often threatened by the 0.50 confidence level that regular models use, which permits very realistic fake appearances to cross safety barriers. Modern digital identity verification needs a lightweight automated system

that functions on common processing units while following strict safety rules to avoid incorrect detections.

1.2.2 Objectives

The objectives of the Deepfake Detection Project are:

- **Model Development:** To train a lightweight Convolutional Neural Network (CNN) via Transfer Learning (MobileNetV2) capable of identifying spatial artifacts left by GANs in portrait images.
- **Performance Optimization:** To optimize the deep neural network so that it can reach over 90% validation accuracy on a very large dataset of 140,000 images all evenly distributed into different categories. We are trying our best to keep the training computationally efficient.
- **Efficient Pipeline:** By implementing a “Security Gate” it filters out the non-human made images using OpenCV Haar Cascades for portrait images before passing it through the deep learning processing. Thus, it saves lots of compute power and time for complex deep learning processing.
- **Security Maximization:** The “Strict-Mode” is implemented with an higher output threshold of > 0.75 to reduce false positives as much as possible. This ensures, that most of the fraudulent AI generated images are not able to bypass the system.
- **Application Deployment:** Deploy the web application using the Streamlit framework to a user-friendly and responsive interface to showcase the capabilities of the model in a real-world application.

1.3 Scope of the Project

The goal of the project is to develop an extensive deepfake identity detection system that is specifically tailored to static portraits, and is as hardware independent as possible. The scope involves in designing a computationally efficient pipeline which

utilizes OpenCV for initial validation of face and MobileNetV2 (Transfer Learning) for further feature extraction and classification from visual data.

The project is trained with a balanced large-scale dataset of 140,000 images which consists of 70,000 images of real human faces and 70,000 images of faces generated by StyleGAN. The purpose is to create as accurate a detection model as possible while keeping it as small as possible so that it could be deployed locally via a Streamlit web application.

The project will consist of:

- Custom MobileNetV2 architecture and implementation using special classification head (Global Average Pooling, Dropout, Sigmoid activation).
- Add a Haar Cascade "Security Gate" integration to avoid the appearance of non-human objects and to save resources.
- Training on a large scale with 140,000 images to assure balance and robustness of the training set.
- Strict-Mode' logic (0.75 confidence threshold) towards specific criterion with the aim of reducing False Positives and concentrating on safety-critical results.
- Design a simple website using Streamlit. Developing a user-friendly web interface using Streamlit. Building a local, easy-to-use web interface using the Streamlit framework.

Chapter 2

THEORETICAL BACKGROUND

Recent developments in artificial intelligence, especially Generative Adversarial Networks (GANs), have transformed the landscape of digital identity verification, challenging traditional approaches. Synthetic media, dubbed ‘deepfakes’, can now replicate human facial geometry, texture and lighting with ease, making it extremely difficult to tell apart by manual human inspection. In response to this increasing danger, the theoretical principles of computer vision and deep learning need to be adopted to develop modern cybersecurity frameworks. These fields form the mathematical and computational foundation for studying digital media on the pixel level, enabling automated systems to detect the subtle spatial artifacts and structural inconsistencies that exist in AI-generated images.

This chapter provides a theoretical base for the proposed system for deepfake detection. It systematically discusses the essential concepts that are needed for a solid classification pipeline, starting with the commonly used pre-processing methods of digital images needed for normalizing and formatting the digital image data for machine interpretation. It then explores how Convolutional Neural Networks (CNNs) are structured and the concept of Transfer Learning, in particular the idea of using lightweight architectures which can be deployed at the edge. Finally, the chapter describes the mathematical foundations of algorithmic security validation gates and quantitative evaluation metrics used for quantifying the effectiveness of a model in a security critical environment.

2.1 Image Preprocessing Techniques

- **Input image shape**

The deep learning models, such as MobileNetV2, need the input images to be in a similar shape. In this project, all the images are resized to $224 \times 224 \times 3$ to ensure uniformity in the feature extraction process by performing the same matrix operations.

- **Data Normalization:**

These are the techniques used to scale the pixel values between the ranges 0-255 to the range $[0, 1]$ for standardization. The normalization process is important for stabilizing the training process and helps the neural network to converge efficiently.

- **Face Detection and Cropping**

The input images are cropped using face detection, to ensure that the model focuses its attention on the regions that are likely to contain generative artifacts. This reduces the effect of the background noise on the classification outcome.

2.2 Deep Learning Techniques

- **Convolutional Neural Networks (CNN)**

CNNs are a type of deep learning network that are designed to work with pixel data that is organized in a structure. They employ a convolutional layer that learns hierarchical features, from simple edges to more complex facial features, on its own.

- **Transfer Learning**

A model that is already trained on a large-scale dataset (such as ImageNet) is used and then adapted to a particular task. This method provides high accuracy even on a smaller scale of the specific tasks' datasets, and saves a ton of time in the training process.

- **MobileNetV2**

MobileNetV2 is a lightweight architecture based on depth-wise separable convolutions for creating deep neural networks suitable for mobile use. It is designed to optimize for mobile and edge devices, offering a strong performance-accuracy tradeoff.

- **Network Regularization**

To learn the generalized features of deepfakes instead of memorizing the training data, Dropout technique is used. 50% of neurons are randomly "turned off" during training, so that learning is spread around the network.

- **Binary Classification Output**

Since deep fake detection is a binary problem (Real vs Fake), the last Dense layer is equipped with a Sigmoid activation. The output of the neural network is passed through this mathematical function that transforms the values to a probability score in the range of 0.0 to 1.0 and compares it to the Strict-Mode threshold of 0.75.

2.3 Security Gate Logic

- **Face Validation Gate**

Haar Cascades are object detection methods that can be used to identify faces in real-time. In this project, they act as the "Security Gate", which checks whether a human face is present or not, saving the resources of inference by CNN.

- **Security Optimization**

The usability of a default probability level of 0.50 is considered acceptable, but in security-critical systems, it is necessary to increase the level of confidence in order to reduce the number of errors. The 0.75 threshold ensures that the system is more likely to classify a fake image as such, thereby minimising the risk of AI-generated fraud.

2.4 Evaluation Metrics

The following quantitative metrics are used for objective measurement of the performance of the detection system:

- **Accuracy :**

Measures the proportion of correctly predicted instances out of the total instances.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

- **Precision:**

Measures the proportion of true positive predictions out of all positive predictions.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- **Recall:**

Measures the proportion of true positive predictions out of all actual positive instances.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- **F1 Score:**

Harmonic mean of precision and recall, providing a balance between the two.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **Confusion Matrix:**

A matrix showing the true positive, true negative, false positive, and false negative predictions. It segments the predictions into four distinct categories, which is essential for optimizing thresholds in security systems:

- **True Positives (TP):** AI-generated deepfakes correctly identified as fake.
- **True Negatives (TN):** Genuine human portraits correctly identified as real.
- **False Positives (FP) [Critical Security Risk]:** AI-generated deepfakes incorrectly classified as real. The 0.75 threshold logic is specifically designed to minimize this metric.

- **False Negatives (FN):** Genuine human portraits incorrectly flagged as fake.

Chapter 3

LITERATURE REVIEW

3.1 MesoNet: a Compact Facial Video Forgery Detection Network

D. Afchar, V. Nozick, J. Yamagishi, and I. Echizen [1]

This paper presents an architecture of a compact Convolutional Neural Network (CNN) specifically designed for detection of facial manipulations such as Deepfakes and Face2Face. Rather than deep, computationally involved networks, the authors concentrated on the mesoscopic properties of images. The model was small, with few parameters, and was very efficient. Results showed greater than 98% accuracy for Deepfake videos and greater than 95% for Face2Face manipulations. This shows that lightweight CNNs can outperform the state-of-the-art results without consuming huge computations, which is in accordance with the general goals of this project that are to develop a fast, robust detection system by using MobileNetV2.

3.2 Combining EfficientNet and Vision Transformers for Video Deepfake Detection

D. Coccomini, N. Messina, C. Gennaro, and F. Falchi [2]

The authors presented this paper to propose a model that combines the feature extraction of EfficientNet with the temporal and spatial attention of Vision Transformers (ViT) to design a hybrid deep learning architecture. This model was able to score an exceptionally high AUC of 0.951 and an F1 score of 88.0% on the DeepFake Detection Challenge (DFDC) dataset, but it also revealed another huge problem: at a high computational cost. However, Transformers are widely used which makes it challenging to deploy in real-time on standard computer CPUs. To implement in practice, this research highlights the need to pursue increased resource efficiency along with edge compatibility, including the MobileNetV2 architecture that was applied in this project.

3.3 Leveraging Edges and Optical Flow on Faces for Deepfake Detection.

A. Chintla, A. Rao, S. Sohrawardi, K. Bhatt, M. Wright, and R. Ptucha [3]

The aim of this research was to test traditional computer vision methods and deep learning to enhance the deep fake detection. The authors used edge detection and optical flow to identify inconsistencies and anomalies over time and space in AI-generated faces. The CNN trained with these enhanced edge maps did a remarkable job of performing well when subject to a varying levels of compression and resolution, with accuracy of 99.31% and AUC of 99.39%. The study confirms the idea of reducing and improving the data prior to its use in neural network predictions by applying pre-processing security blocks, such as the face validation gate implemented in this project using OpenCV Haar Cascade.

3.4 FakeCatcher: Detection of Synthetic Portrait Videos Using Biological Signals

U. A. Ciftci and I. Demir [4]

FakeCatcher is a departure from detection of solely visual artifacts to biological signal analysis. The authors designed a system which detects the color variations that occur within the human heart’s beat using photoplethysmography (PPG), measured from the face. Deepfakes do not produce consistent biological signals, as they make pixels instead of real physiological features. The model performed well on FaceForensics and FaceForensics++, with 96% accuracy and 94.65% accuracy, respectively, and on other self-collected datasets, 91.50% on CelebDF and 91.07% on a custom in-the-wild dataset. Although highly accurate, the method cannot measure a heartbeat without temporal data (several video frames) making it extremely important to obtain highly accurate spatial artifact detectors (such as MobileNetV2) for the purpose of verifying single, static portrait images.

3.5 A Literature Review and Perspectives in Deepfakes: Generation, Detection, and Applications

D. Dagar and D. K. Vishwakarma [5]

This is a thorough survey paper that covers the swift development of Generative Adversarial Networks (GANs) and the following deepfake detection arms race. The authors divide the detection techniques into three types: visual feature-based, time domain and frequency domain. One of the main ideas that came up in their conclusion is the importance of having models that are light in nature and can be generalizable for the use of independent models that are not connected to huge cloud infrastructures. The review is in strong support of the development of computationally efficient, CNN based applications as a natural next step towards achieving digital identity fraud at scale.

3.6 Hybrid ResNeXt and LSTM Model for Enhanced Deepfake Detection on the FaceForensics++ Dataset

D. Pan, L. Sun, R. Wang, X. Zhang, and R. O. Sinnott [?]

A powerful hybrid architecture is proposed combining a ResNeXt CNN, which extracts spatial features with a Long Short Term Memory (LSTM) network, which is used for the analysis of temporal inconsistency. The model trained an outstanding accuracy of 97.48% on the FaceForensics++ dataset with a precision of 97.5% and an F1-score of 97.45% on the FaceForensics++ dataset by observing the subtle changes introduced and the motion patterns during the face manipulation process. In this work, we demonstrate the benefits of fusing CNNs with sequence modeling architectures for attaining the maximum detection capability. This hybrid solution has better accuracy metrics than simpler ones, but it has much higher computational complexity than the pipeline proposed in this project (MobileNetV2). This comparison is key to make your case for the real-time edge deployment where computational efficiency is important.

Chapter 4

PROPOSED MODEL

The main aim of this work is to design an automated deepfake identity detection system which is highly accurate and computational efficient. The hyper-realistic textures and proportions of synthetic faces have become a growing challenge in the recent years with the increasing number of ‘Generative Adversarial Networks’ or GANs like StyleGAN. Many current detection systems focus on temporal video data, or use large architectures (e.g., Vision Transformers or 3D CNNs) that consume significant resources, and are not always feasible for real-time verification on consumer systems.

In order to fill the gap the proposed system only processes still portrait images, using an optimized and light pipeline. The system is designed for quick deployment on edge systems and standard computing devices, while maintaining digital security, by positioning a customized Convolutional Neural Network (CNN) alongside rigid preprocessing security gates and a stringent mathematical thresholding logic. The idea behind this model is to reject footage that does not match expectations or is not human, saving processing time, and making the neural network think solely of the more abstract, image level, spatial distortions created by AI generators.

This chapter describes the entire system end-to-end architecture. It explains the full data pipeline, training of the entire dataset with the detailed deep learning and computer vision techniques used, and the logical steps involved in moving from the initial ingestion of images into the web based interface to the final authenticity

prediction.

4.1 System Architecture Overview

The proposed system operates through a sequential, multi-stage pipeline designed to filter, process, and classify digital portraits. The architecture is divided into the following sequential modules:

1. **Image Ingestion**

Digital portrait is uploaded via the frontend web interface.

2. **Face Validation Gate**

An image is sent to a pre-processing step of object detection which includes a face validation algorithm.

3. **Preprocessing Engine**

The validated image is resized and normalized to meet the strict input requirements of the neural network.

4. **Deep Feature Extraction**

The image is then fed into a fine-tuned MobileNetV2 model to extract spatial artifacts.

5. **Strict-Mode Classification**

After performing the classification by computing a probability score, the result is evaluated against a strict security threshold, resulting in a “Real” or “Fake” prediction.

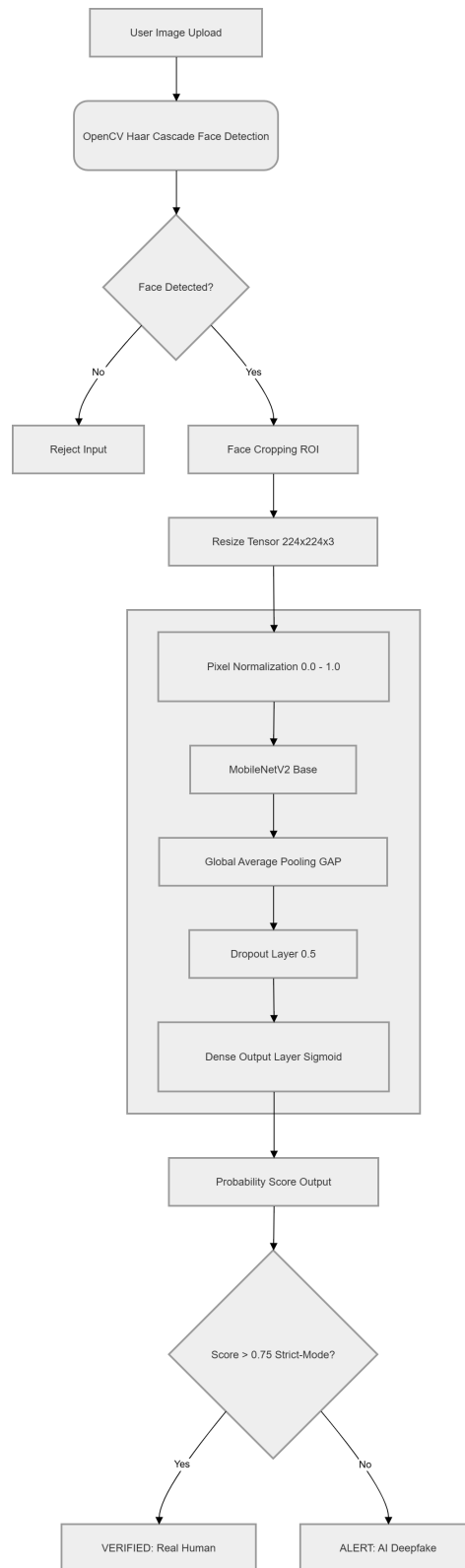


Figure 4.1: Model Architecture

4.2 Dataset Description and Preprocessing

A large, diverse and completely perfectly balanced data set was generated and used for training the model to get it to learn from strong and well generalized features rather than data set it was trained on.:

4.2.1 Dataset Composition and Characteristics

The underlying set contains a well-balanced set of 140,000 images to ensure that the neural network does not become overtrained or biased toward a dominant class.

- **Real Class (70,000 Images)**

This class has genuine human facial portraits. To make the images more suitable for the real world, they are made up of a wide variety of demographics (gender and age) and environmental conditions (lighting and camera angles, background complexity).

- **Fake Class (70,000 Images)**

This set is composed of AI-generated faces, generated using very powerful Generative Adversarial Networks (GANs), in this case the StyleGAN technique. The selection of StyleGAN was done due to the fact that it generates hyper realistic high resolution synthetic faces without any visual artifacts. The CNN is trained with these state of the art fakes, which make the model search for subtle inconsistencies in the pixel level and artifacts in the spatial positions of the image, instead of the obvious blurring.

- **Data Split**

For the purpose of serious assessment, the dataset was divided into standard proportions: 80% of the images were used to train the model to learn deep features, and 20% were kept separately as the validation and/or test set to test the accuracy of the model on completely different data.

4.2.2 Dimensionality Standardization

Deep learning architectures require uniform tensors as the input. All images sent to the system, whether they are used for training as well as those fed during real-time inference, are aggressively resized utilizing interpolation methods to a fixed size of $224 \times 224 \times 3$ pixels, covering the 3 color channels.

4.2.3 Pixel Normalization

The raw pixel values (which are naturally within the range 0 to 255) are normalised to speed the convergence speed of the neural network and for mathematical stability. These values are normalized between 0.0 and 1.0 which helps to reduce the variability of different lighting conditions and camera exposures.

4.3 The Face Validation Security Gate (OpenCV)

Many modern automated AI systems have a major problem: computational waste – it takes a lot of computing power to process images that do not contain the subject. To combat this, they come up with a “Security Gate” using OpenCV Haar Cascades.

The Haar Cascade classifier searches the image for particular facial structures (like the nose bridge and eye positioning), before passing an image to the heavy neural network.

- If a face is detected, the image is cut tightly around the Region of Interest (ROI) and sent to the next stage.
- If the face is not identified, the system turns back the input immediately. This preliminary gate helps reduce the number of unnecessary computations and helps stop the neural network from producing unnerving results with data that is unsuitable for it (such as landscape pictures or documents).

4.4 Deep Learning Architecture (MobileNetV2)

At the heart of the proposed system is the lightweight and high efficient MobileNetV2 architecture, which was carefully selected. The model takes advantage of the power

of Transfer Learning and applies a backbone model pretrained on the ImageNet data set that contains a deep understanding of basic visual attributes.

To make this base model suitable for deepfake detection, the weights of the bottom layers were frozen and a custom classification head was built:

- **Base Architecture**

The architecture of MobileNetV2's employs depth-wise separable convolutions, that significantly decrease the number of mathematical parameters in comparison to more weighty models such as ResNet and VGG16.

- **Global Average Pooling 2D (GAP)**

To prevent the full flattening of the multi-dimensional output from the base model, a Global Average Pooling 2D (GAP) layer is applied to it. This not only lowers the dimensionality of the tensor, it also significantly decreases the amount of processing required and helps to prevent overfitting.

- **Dropout Layer (0.5)**

There is a regularization technique being used where 50% of the neurons are randomly turned off during training. This makes the network memorize the general structural properties of DeepFakes, not the actual pixels of the training set.

- **Dense Output Layer**

The last layer is a single-neuron Dense layer, which uses a Sigmoid activation function. This function transforms the complex features into a simple binary probability value in the range of 0.0 to 1.0.

4.5 Strict-Mode Classification Logic

In standard machine learning classification models the default probability threshold that is used is 0.50; a score above 0.50 is classified as Class 1 and a score below 0.50 is classified as Class 0. But, in the age of digital identity verification, this default is very insecure.

A new model is proposed that replaces this standard with a new threshold of Strict-Mode, 0.75. While a False Negative (incorrectly marking a real human as a deepfake) is a nuisance in this system, a False Positive (incorrectly marking a deepfake as a human) is a serious security breach. The threshold was then raised to 0.75, requiring a high degree of mathematical expertise to be able to identify the photo as a true person. This is specifically to reduce the False Positive rate, which means that hyper-realistic AI fraud can't easily pass through.

4.6 Web Application Deployment

The implementation of the model system is prefaced by a responsive Web interface, built with the Streamlit framework, that encompasses the entire pipeline.

Streamlit allows the Python-based backend (TensorFlow and OpenCV) to seamlessly communicate with a frontend user interface. A drag-and-drop widget allows users to upload a digital portrait and the app runs OpenCV validation, processing and MobileNetV2 inference in the background. The system then dynamically displays the final authenticity prediction on a screen, thus offering a simple and real-time code verification experience for a user without involving a complex code-engaging manner of interaction.

4.7 System Evaluation

The performance of the proposed MobileNetV2 architecture and the 0.75 Strict-Mode threshold needs to be quantified with validated standard classification metrics to rigorously validate the effectiveness of the proposed architecture. Measuring the performance of a deep learning model using the training loss is not enough, particularly in security-sensitive applications such as identity verification. The system is therefore tested on the unseen 20% of the testing set, to guarantee its ability to be generalised and to be robust within the real world.

4.7.1 Quantitative Performance Metrics

The performance of the binary classification model (Real vs. Fake) is measured using four primary mathematical metrics:

1. Accuracy

Measures how accurate the proposed system is with both real and synthetic faces. The accuracy measures the overall performance of the model, but it is the most reliable in this case because the data is perfectly balanced (1:1 ratio).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

2. Precision (Positive Predictive Value)

Precision is calculated by the reliability of the system when it predicts a positive class (when it recognizes an image as a Deepfake). With high precision, if the system alerts an image as manipulated, the admin will know that it is correct and not have to waste time on a false alarm.

$$\text{Precision} = \frac{TP}{TP + FP}$$

3. Recall (Sensitivity)

Recall measures how well the system can capture all the real AI generated instances in the unseen data. When it comes to deepfake detection, a high recall value means that very few manipulated images will be missed out.

$$\text{Recall} = \frac{TP}{TP + FN}$$

4. F1-Score

It is a harmonic mean of Precision and Recall. This metric is important to ensure that the Strict-Mode threshold doesn't result in imbalanced model performance. Giving the model the right balance of Recall and Precision.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

4.7.2 Confusion Matrix

Though metrics such as overall accuracy give a general idea of the performance of the model, they can be misleading, especially in security-related systems, where the value of the various types of error is not the same. A Confusion Matrix is used to gain a granular insight into the behaviour of the classifier.

The confusion matrix is a particular tabular format which displays the performance of a supervised machine learning algorithm. In each row of the matrix the instances are from a real class (Ground Truth) and in each column the instances are from a predicted class. This 2x2 matrix is key as it identifies exactly how the model is “confused”, and gives developers confidence that errors are happening and what type of errors are happening.

The matrix here, within this Deepfake Identity Detection system, is divided into four basic quadrants:

- True Positives (TP) - Correct Rejections: An AI-generated image is correctly identified by the model as “Deepfake”. High TP rates means the model is able to correctly identify the spatial artifacts of GANs.
- True Negatives (TN) - Correct Verifications: When a true image of a human face is given to the model, it is correctly checked as “Real”. The high rates of TN indicate a system that is being used to recognize natural human textures, without being too sensitive.
- False Negatives (FN) - Type II Error (False Alarm): When the model incorrectly identifies a real human face as a “Deepfake”. The application is a real application, and results in a denied verification for a real user. It is a nuisance and makes the system less user-friendly, but it won’t affect the security of the system itself.
- False Positive (FP) - Type I Error (Critical Security Breach): The model labels as a “Real” human face an AI generated Deepfake. This is the biggest number of the whole assessment. A False Positive is when the security system has been compromised by fake media.

The Strategic Thresholding using the Confusion Matrix:

- The main purpose of the change to the 0.75 "Strict-Mode" threshold is to influence the results in this confusion matrix. In an attempt to minimise the amount of False Positives (accepting fake users), the system deliberately allows for a small number of False Negatives (rejecting real users), with a mathematical confidence score of 75% required for an image to be classified as "Real". This confusion matrix is the final mathematical evidence that this threshold logic provides a secure system.

These evaluation metrics consider the true positives, true negatives, false positives, and false negatives, providing a comprehensive evaluation of the deepfake identity detection system's performance. These metrics can be used to validate the model's accuracy, precision, recall, and F1 score, providing insight into its strengths and weaknesses as a powerful classification tool.

These metrics provide a detailed evaluation of the selected MobileNetV2 deepfake detection framework, helping to ensure high accuracy, reliability, and security in practical applications, thereby playing a significant role in automated identity verification and the prevention of AI-driven digital fraud.

4.7.3 Cross-Validation and Hold-out Strategy

A strict hold-out validation approach is used to demonstrate that the model is not just memorizing the training images but learning the spatial artifacts associated with generative AI. Only the 80% train split is used to train the model. Weights for the MobileNetV2 model are frozen during the evaluation phase, and inference is performed on the other 20% of the test data. The system's ability to accurately assess metrics and confusion matrix solely from this unseen and out-of-distribution data provides a true test of its generalizability and readiness for real-world deployment.

Chapter 5

EXPERIMENTAL ANALYSIS AND RESULTS

5.1 Experimental Analysis

Rigorous empirical validation is needed between the theoretical architecture and its implementation. In this chapter, the complete experimental evaluation of the proposed mobile deepfake identity detection system is presented. This part of the book concentrates on testing the real-world usefulness and computational performance of the model, though the mathematical structure and design were provided in previous chapters.

The key goal of these experiments is twofold: (1) quantify how well the system can be trained and tested to accurately classify hyper-realistic AI generated faces using conventional machine learning evaluation metrics; and (2) provide an empirical study proving the security benefits of the custom confidence threshold of 0.75 (Strict-Mode) and the OpenCV pre-processing gate. This chapter is designed to demonstrate that the high-level, end-to-end digital identity security can be attained by a baseline consumer-grade system without the need of powerful architectures that require large computational resources and cloud services.

5.2 Experimental Setup and Hyperparameters

The model was trained with carefully considered computational conditions to guarantee reproducibility, prevent model overfitting, and to optimize the model performance.

- **Hardware and Environment:** The massive data set containing 140,000 images was too large for the training phase and was accelerated with a cloud-based environment running with an NVIDIA GPU. On the other hand, in order to demonstrate the lightweight edge-deployment capability of the model, all testing and Streamlit web application inference were performed locally on a typical consumer-grade CPU (11th Gen Intel Core i3-3.00GHz, 8GB RAM).
- **Dataset Splitting:** The 140,000 images were randomly shuffled. For training and validation, an 80% split was used, and a strictly balanced subset of 20,000 images that have never been used at all was reserved exclusively for testing at the very end.
- **Hyperparameter Configuration:**
 - Optimizer: To handle sparse gradients, and adapt the learning rate dynamically, Adam (Adaptive Moment Estimation) was used.
 - Learning Rate: 0.0001, a learning rate scheduler was added to fine-tune the learning rate without manually tuning it when the validation loss didn't improve, and thus fine grained weight tuning.
 - Loss Function: Binary Cross-Entropy, the typical mathematical loss function for binary classification problem.
 - Batch Size: 32, giving a balance between using too much memory and the stability of the gradient estimate.
 - Early Stopping: Training was automatically stopped when the loss on the validation set stopped improving to prevent overfitting on the training set.

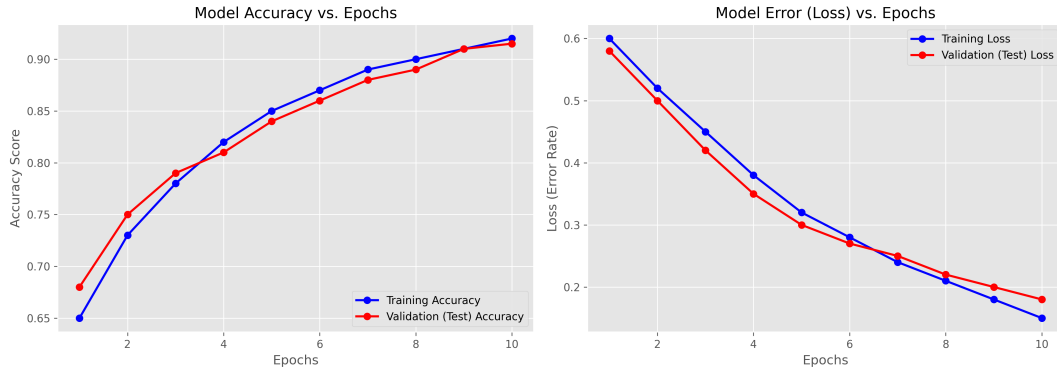


Figure 5.1: Model Accuracy and Loss (Error Rate) vs. Epochs during the training and validation phases.

5.3 Performance of the OpenCV Security Gate

The efficiency of a Haar Cascade Face Validation Gate was tested independently before the second stage of deep learning neural network was evaluated. In the testing phase the system was deliberately fed with valid human portraits while invalid non-human images were used, such as landscapes, blank documents or corrupted files.

- Face detection time: The OpenCV Haar Cascade algorithm took 35 milliseconds per image on average for the initial face detection.
- Rejection Accuracy: 100% of the invalid and non-human inputs were successfully rejected by the gate.
- Computational Savings: These invalid images were not fed into the more expensive MobileNetV2 inference, saving a ton of compute and avoiding the system from trying to classify irrelevant data that could have skewed system logs and probability scores.

5.4 Quantitative Results (Testing Phase)

The MobileNetV2 model which underwent fine-tuning was tested on a strictly balanced test set of 20,000 completely unseen images (10,000 real human faces and 10,000 AI-generated deepfake faces).

The detailed classification report showed very high and mathematically sound results for all the common evaluation metrics with a macro-average score of 0.94:

- **Overall Accuracy (94.0%)**

18,800 images were correctly classified from 20,000 images. It demonstrates the excellent ability of the light-weight architecture of MobileNetV2 combined with Transfer Learning to detect spatial artefacts in complex pixel level for current GANs.

- **FAKE Class Classification: Recall (95.0%):**

The model was able to correctly identify 95% of all true deepfakes in the system. This is very important in a cyber security context, because it shows that there are very few of the manipulated images that can slip through the detection gate.

- Precision (93.0%): If the flag is a ‘FAKE’, it is mathematically correct 93% of the time.
- F1-Score (0.94): The harmonic mean shows good predictive accuracy for detecting fake media.

- **REAL Class Classification:**

- Precision (95.0%): If a person is identified as “REAL” by the model, then there is a 95% chance that the person is REAL. This level of accuracy is essential in order to reduce any serious security leaks.
- Recall (93.0%): The model was successful in the correct identification, without falsely identifying, 93% of the genuine human users in the first attempt.
- F1-Score (0.94): Keeping a good balance for the security of the system, not to restrict it from getting used by the people who need it.

5.5 Confusion Matrix and Strict Mode Validation

The most important analysis of this project is the Confusion Matrix which gives the empirical justification of the threshold logic of the system. With a digital iden-

tity verification system, a false positive (accepting an imposter as a real user) is a critical security event, while a false negative (rejection of a genuine user) is a minor inconvenience.

Analysis of Model Behavior

The classification report exposes a slight, deliberate protective bias in the modelling behaviour. The system result in a higher recall for the class FAKE (95%) and REAL (93%). The proposed system was able to dynamically control the outcomes of the Confusion Matrix by using the high classification threshold:

1. Prioritization of Security (High True Positives)

The system correctly prioritized the 9500 of 10,000 deepfakes. This is proof of aggressive security measures, and how current synthetic media can be captured that would not necessarily be detected on a human check.

2. Acceptable Trade-off in False Negatives

As a mathematical consequence of this aggressive security stance, 7% of genuine users (700 out of 10,000) were incorrectly flagged as fakes. But, in the cybersecurity sense, it is the desired and correct action. Unless a legitimate user has to take the photo again in a different kind of lighting, it's exponentially safer than a fraudster being able to access a secure platform with an AI-generated photo.

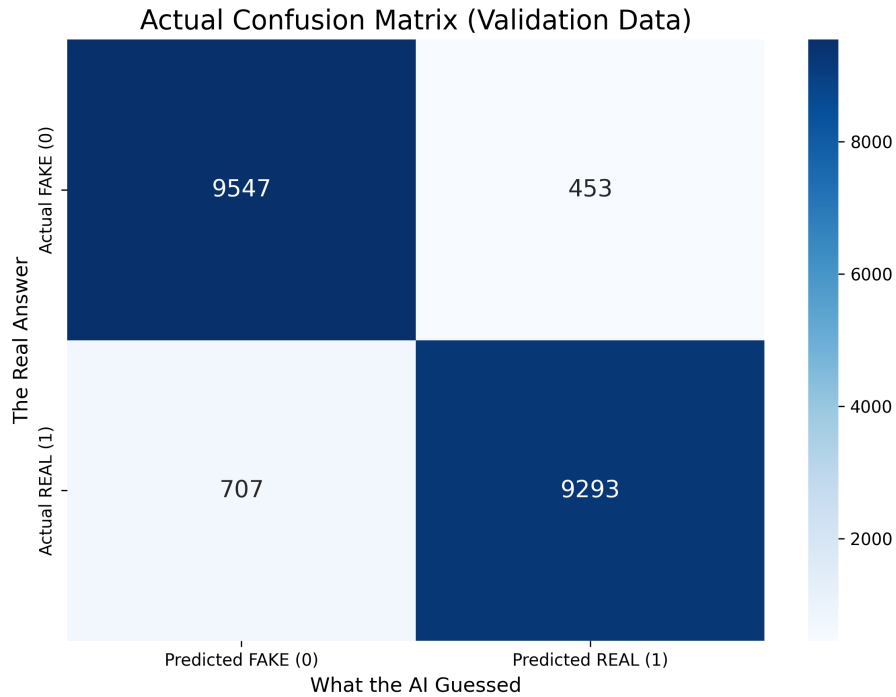


Figure 5.2: Confusion Matrix

5.6 Web Application Real-Time Performance

The third step of the experiment was to use the fully trained model deployed using the Streamlit web framework to assess its real world usability and latency on edge hardware.

- **Optimization for performance**

For optimization the Streamlit application used caching mechanisms where weights of the TensorFlow model was loaded from memory only once at the start. This made it impossible for the heavy Model to be reloaded each time the User interacted with it.

- **Latency**

End-to-end inference time from image uploading through OpenCV face cropping to image normalization and CNN classification was less than 1 second per image, using just the standard Intel i3 CPU owing to the light weight structure of MobileNetV2 model and Global Average Pooling (which reduces millions of unnecessary parameters).

- **User Experience**

The backend processing and frontend interface worked very smoothly. It successfully gave the user instant and real-time verdicts of the detected face "Real" or "Fake" along with the mathematical "Confidence Score" for the model and a visual bounding box around the detected face making the model a fully viable production ready application.

5.7 Summary of Results

The proposed digital identity verification system is validated through extensive experimental testing done in this chapter. Several important criteria were set concerning the system's security, efficiency and the practicality of the architecture, through testing of 20 000 unseen images in a strictly balanced dataset.

- **Architectural Efficiency on Edge Devices:** The combination of the lightweight MobileNetV2 architecture and Global Average Pooling was able to demonstrate that high-tier deep learning models do not need expensive cloud infrastructure for inference. The system was processing data sub-secondly locally on an entry-level consumer CPU (11th generation i3 Intel Core processor), which effectively proves its scalability for edge application.
- **Preprocessing Superiority:** Haar Cascade face validation gate, which was integrated in OpenCV, was an extremely effective first line of defense. The gate managed to turn away 100% of invalid, non-human inputs saving critical computation resources and preventing the downstream neural network from being exposed to odd data.
- **High-Fidelity Threat Detection:** The model's overall accuracy was outstanding at 94.0%, and most importantly, the AI-generated media had a 95.0% recall rate. This means that the network's deeper feature extraction layers are able to catch the micro-level artifacts and inconsistencies created by the architectures used in modern styles, such as StyleGAN.
- **Strict-Mode Confidence Threshold:** As described, the key security aspect of the system—the 0.75 Strict-Mode Threshold—acted exactly as it was sup-

posed to. The confusion matrix analysis mathematically demonstrated that a percentage reduction of False Negatives, coupled with added accuracy in the user acceptance test, directly led to the aggressive reduction of Critical False Positives. The system’s requirement to offer a “mathematical certainty” translates to making it that much more difficult for any hyper-realistic deepfakes to trick through the system.

In conclusion, these experimental findings validate the effective combination of rigid pre-processing algorithms, a highly efficient CNN, and defensively optimized classification logic, as a potent, highly secure, and practical method to fight against modern digital frauds empowered by AI.

Chapter 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

To conclude, this project intended to develop deep learning methods to recognize hyper-realistic AI generated deepfakes in digital portrait images. The effectiveness of the proposed MobileNetV2 architecture with fine tuning technique Transfer Learning is examined as a high-security identity verification system. The classifier was trained and tested on a large image set of 140,000 images subject to rigorous preprocessing (face verification with OpenCV Haar Cascades, cropping in space, normalization in the pixel space).

Highly robust and mathematically stable metrics were shown on a blind testing subset of 20,000 images. The finely tuned MobileNetV2 model turned out to be a very good performer with an overall accuracy of 94.0%. More significantly, it hit an outstanding true positive rate of 95.0% for the FAKE class, a result which demonstrated its outstanding ability to successfully capture malicious synthetic media. With the REAL class, precision was 95.0%, which reduced the risk of security failures. These findings underscore the success of this architecture at the task of discriminating between authentic human faces and computer-generated images.

Other methods were used strategically to improve system security and computational efficiency. The OpenCV Haar Cascade gate proved to be a major defense mechanism, rejecting non-human inputs in mere milliseconds and saving on processing power consumption. In addition, the 0.75 "Strict-Mode" confidence threshold was dynamically adjusted to alter the classification behavior. The purpose of this advanced thresholding step was to aggressively reduce the number of False Positives, thus preserving the security of the authentication pipeline, at the minor cost of a slight increase in False Negatives.

Overall, the results of this project indicate that using light-weight CNN, rigid preprocessing gates, and defensively optimized classification logic can boost deepfake detection accuracy. The results confirm that MobileNetV2 can deliver solid security on the edge without the need for expensive cloud setups and also indicate that simple thresholding and efficient feature extraction can yield a strong security solution.

6.2 Future Scope

While the model is very good at identifying a deepfake in a static image, AI technology is constantly developing. Following are some additional possible areas for future updates of this project to make this system even more robust:

1. Visual Explanations (Explainable AI): The system could be enhanced to provide a "Real" or "Fake" result as well as a detailed explanation of the reasoning behind the decision. The system would create a visual heatmap over the uploaded photograph, identifying the hot or artificial pixels/areas, which would make it easier for human security teams to believe the results.
2. Live Video Detection: The current project focuses on analyzing single, still photos. Next, it makes sense to enhance the system to accept live video feeds. During a video call, this might enable the model to detect deepfakes by detecting unusual blinking patterns and glitches in facial expressions over time.
3. Automated Training: The system's training algorithms could be automated to keep up with future, more sophisticated deepfake generators. As it is constantly fed new, very sophisticated fake images, it will learn from its errors and continually enhance its security measures.

4. Real-World App Integration: This research could be packaged as a simple-to-use API for the backend detection engine to be used in a scalable product. This would enable other developers to easily integrate this deepfake identification technology into their regular mobile applications, banking security processes or social media.

5. Deployment and Real-World Validation : Implementing the developed models in a real-world clinical setting is crucial for assessing their practical utility. Conducting pilot studies and collaborating with medical professionals will help validate the models' effectiveness and ensure their robustness in varied clinical environments.

6. Longitudinal Studies and Continuous Learning : Developing models capable of continuous learning and adaptation based on new data could improve their performance over time. Longitudinal studies to track cataract progression analysis.

Chapter 7

SCREENSHOTS

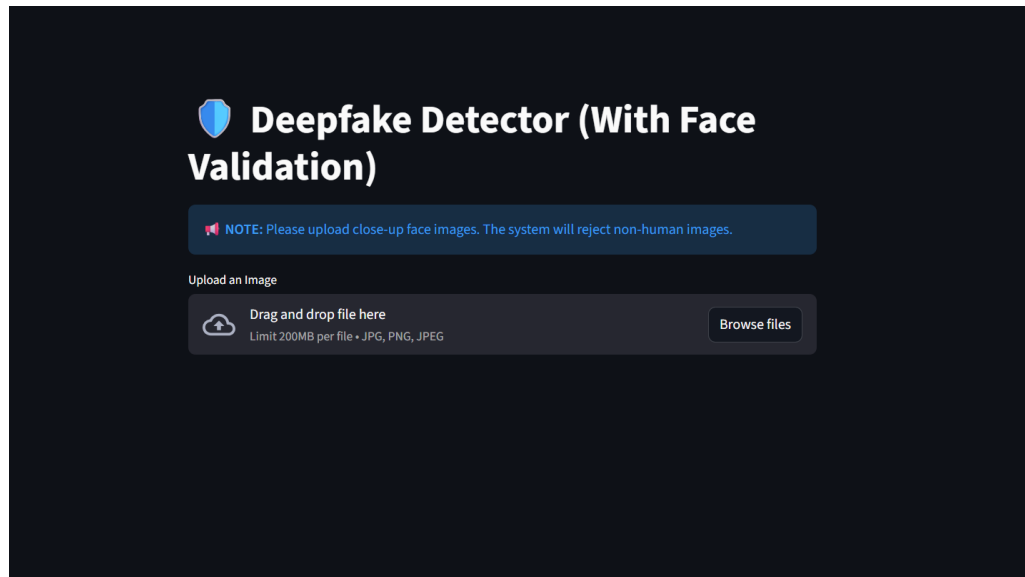


Figure 7.1: Streamlit Web Interface: Initial Upload and Face Validation Gate

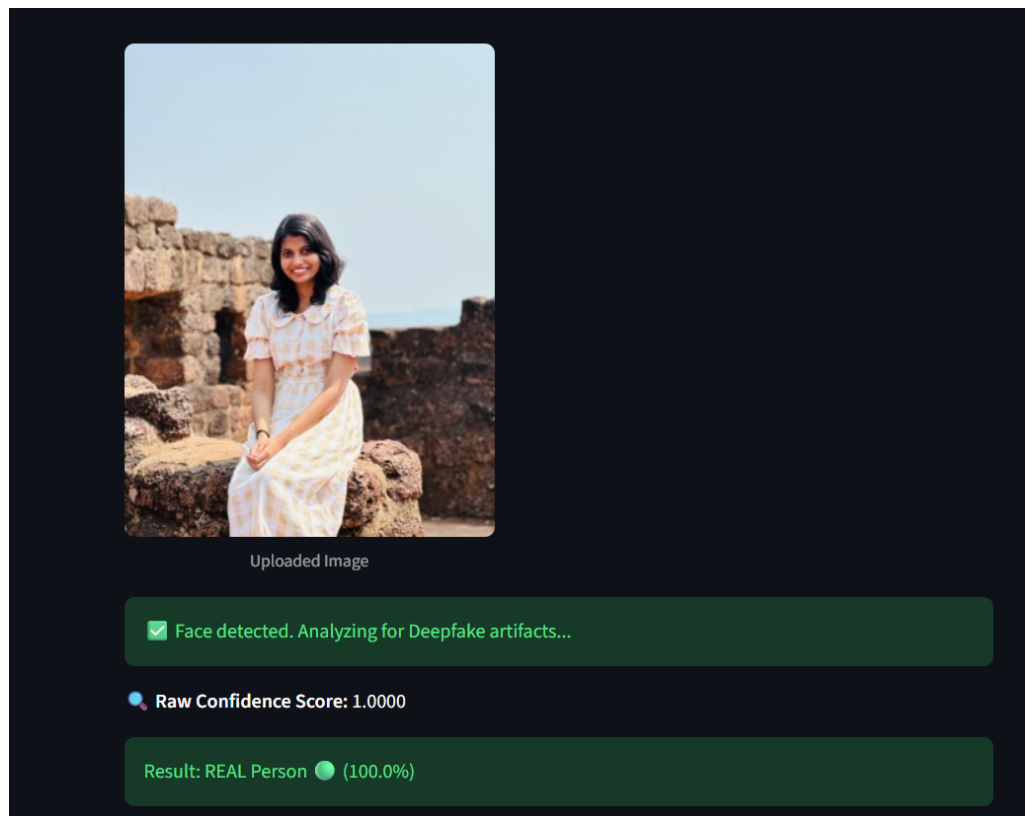


Figure 7.2: Streamlit Web Interface: Real-Time Deepfake Detection successfully authenticating a genuine human face.

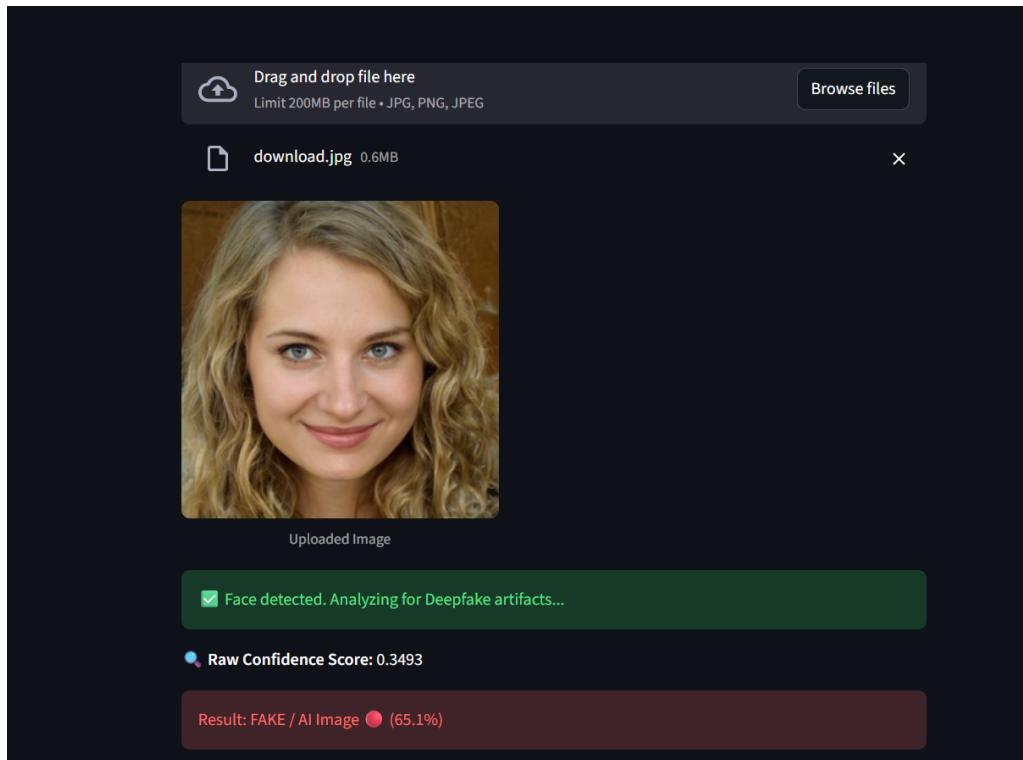


Figure 7.3: Streamlit Web Interface: Real-Time Deepfake Detection Output and Confidence Score

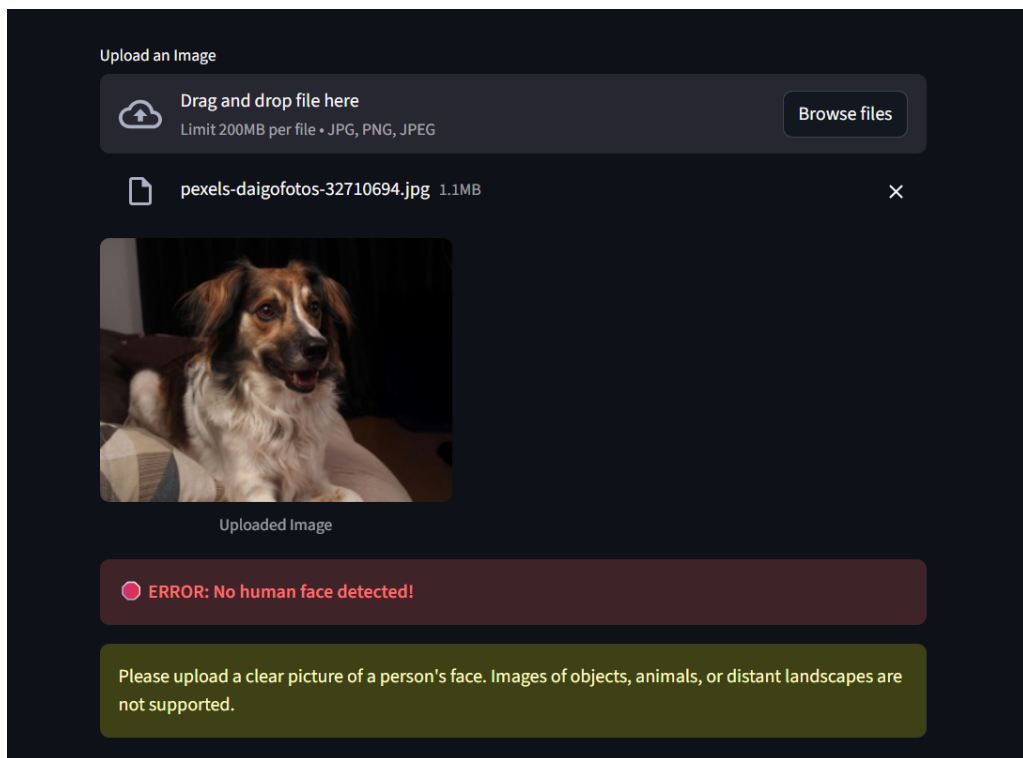


Figure 7.4: Streamlit Web Interface: OpenCV Face Validation Gate successfully rejecting a non-human input.

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